

B.Tech. Degree II Semester Regular and I Semester Supplementary Examination May 2016

IT/CS/EC/CE/EE/ME/SE 1104 B / 1204 A BASIC ELECTRICAL ENGINEERING
(2015 scheme)

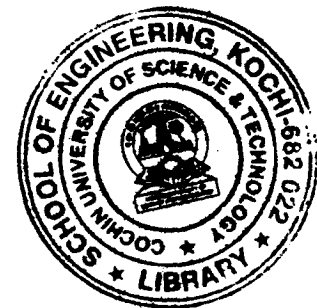
Time : 3 Hours

Maximum Marks : 60

PART A(Answer *ALL* questions)

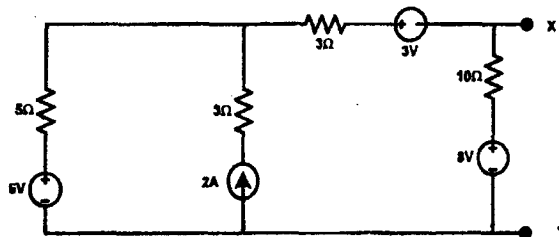
(10 × 2 = 20)

- I.
- State and illustrate Kirchhoff's laws.
 - A capacitor of $47 \mu F$ stores 20 mj of energy. Obtain the amount of charge stored in it.
 - State Coulomb's law.
 - Compare a magnetic circuit with an electric circuit.
 - Explain the working of a Galvanometer, with a neat diagram.
 - The current flowing in a circuit is, $i(t) = 25 \sin(\omega t - 30^\circ)$, when a voltage of $v(t) = 100 \sin(\omega t)$ is applied. Find the active and reactive power.
 - A resistance of 50Ω , inductance of 20 mH, and a capacitance of $70 \mu F$ is connected in series. Find out the total impedance and power factor, if the supply frequency is 50 Hz.
 - Compare between series and parallel resonance for an a.c. circuit.
 - Derive the e.m.f equation of a single phase transformer.
 - Briefly explain the working of an induction motor.

**PART B**

(4 × 10 = 40)

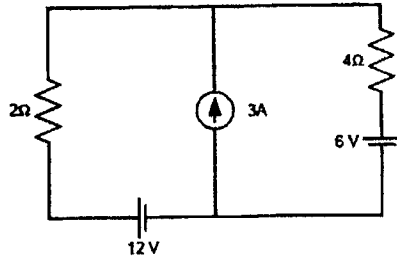
- II.
- State Thevenin's theorem. (3)
 - Obtain Thevenin's equivalent for the circuit given below, across x-y terminals. (7)



OR

(P.T.O.)

- III. (a) State superposition theorem. (3)
 (b) Find the current through the 2Ω resistor of the network given below using superposition theorem. (7)

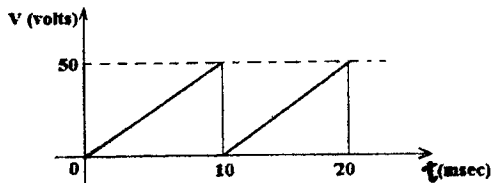


- IV. (a) State Faraday's laws of electromagnetic induction. (3)
 (b) Explain the working of dynamometer type Watt meter, with a neat diagram. (7)

OR

- V. An air gap of 0.2 cm is cut across a steel ring of circular cross-section of area 20 cm^2 . The average length of the flux path around the ring is 4 m. What is the m.m.f required to establish a flux of 2 mWb in the air gap. Relative permeability of steel is 1000. Neglect magnetic leakage and fringing. (10)

- VI. For the periodic wave form given below, find: (10)
 (i) RMS value.
 (ii) Average value.
 (iii) Form factor.
 (iv) Peak factor.



OR

- VII (a) Derive the relation between the line and phase values of voltage and current in a balanced three phase star connection with the help of phasor diagram. (5)
 (b) Two circuits, the impedances of which are given by $Z_1 = (15 + j10)\Omega$ and $Z_2 = (8 - j6)\Omega$ are connected in parallel. If the total current supplied is 20 A, what is the power taken by each branch. (5)

- VIII. What are the different types of D.C. generator? Explain? (10)

OR

- IX. Explain the components of thermal power plant, with the help of a neat diagram. (10)